KKT and Characterizations of Optimality

$\min: f_0(x)$

$\text{st}: f_i(x) \le 0$ primal: x

$h_j(x) = 0$ dual: $\lambda \ge 0$, ν

$(\hat{x}, (\hat{\lambda}, \hat{\nu}))$ satisfy the KKT conditions:

- $\hat{x}$ primal feasible: $f_i(\hat{x}) \le 0$, $h_j(\hat{x}) = 0$
- λ, ν dual feasible: $\lambda \ge 0$
- $\lambda_i f_i(\hat{x}) = 0 \quad \forall i$: complementary slackness
- $\nabla f_0(\hat{x}) + \sum \lambda_i \nabla f_i(\hat{x}) + \sum \nu_j \nabla h_j(\hat{x}) = 0$: stationarity condition.

<u>Thm</u>: If strong Duality holds, then $(\hat{x}, (\hat{\lambda}, \hat{\nu}))$ is optimal <u>iff</u> it satisfies KKT conditions.

Unconstrained convex optimization

$$\min: f_0(x)$$

No constraints. No dual variables.

No dual feasibility. No primal feasibility.

No CS.

$\hat{x}$ is optimal iff

$$\boxed{\nabla f_0(\hat{x}) = 0}$$

Geometric Viewpoint

" $f_0'(\hat{x}) = 0$ " – " $\nabla f_0(\hat{x}) = 0$ "

min: $f_0(x)$

$\hat{x}$ optimal? optimal when we <u>cannot</u> improve the solution <u>locally</u>.

$$f_0(\hat{x} + \varepsilon d) \geq f_0(\hat{x})$$

ε small: $f_0(\hat{x} + \varepsilon d) \simeq \cancel{f_0(\hat{x})} + \varepsilon \cdot \nabla f_0(\hat{x})^T d \geq \cancel{f_0(\hat{x})}$

$$\nabla f_0(\hat{x})^T d \geq 0 \quad \underline{\underline{\forall d}} \quad \Rightarrow \nabla f_0(\hat{x}) = 0$$

Constrained convex optimization — 1/4

min: $f_0(x)$

st: $f_i(x) \le 0$

$h_j(x) = 0$

$\hat{x}$ opt. if it is feasible and all <u>allowable</u> dir. make life worse — do not decrease $f(x)$.

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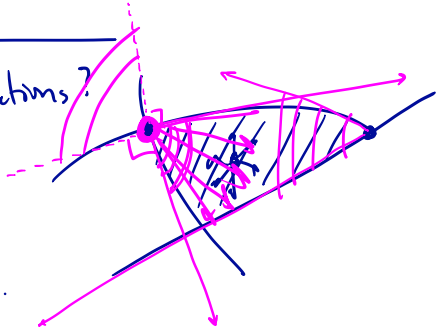
① What are allowable directions?



② We do not decrease $f_0(\hat{x})$ if we move in a dir. d that makes $\geq 90^\circ$ with $-\nabla f_0(\hat{x})$

Constrained convex optimization — 2/4

What are the allowable directions at a feasible pt $\hat{x}$?

d "allowable" if $\hat{x} + \varepsilon d$ is feasible for $\varepsilon > 0$ small enough.

$$\boxed{\begin{array}{c} f_i(x) \le 0 \\ \downarrow \\ h_j(x) = 0 \end{array}}$$

$f_i(\hat{x} + \varepsilon d) \le 0$: if $f_i(\hat{x}) < 0$ then d can be anything.

if $f_i(\hat{x}) = 0$

$$f_i(\hat{x} + \varepsilon d) = \cancel{f_i(\hat{x})}^{\,0} + \varepsilon \boxed{\nabla f_i(\hat{x})^T d \le 0} \quad \forall\, i \text{ that is binding.}$$

$h_j(\hat{x} + \varepsilon d) = 0$: $\nabla h_j(\hat{x})^T d = 0$

Constrained convex optimization — 3/4

Allowable dir: d : $\nabla f_i(\hat{x})^T d \le 0$ $\forall$ constraints that are <u>binding</u> at $\hat{x}$

$\nabla h_j(\hat{x})^T d = 0$ $\forall j$.

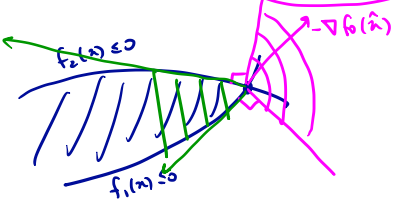
$\hat{x}$ <u>optimal</u> if all allowable dir d <u>make $\ge 90^\circ$</u> with $-\nabla f_0(\hat{x})$

This will happen if:

$$\boxed{-\nabla f_0(\hat{x}) = \sum_{i:\text{binding}} \lambda_i \nabla f_i(\hat{x})} + \sum \nu_j \nabla h_j(\hat{x})$$

$\lambda_i \ge 0$

leave out of illustration



This is the stationarity condition:

$$\nabla f_0(\hat{x}) + \sum \lambda_i \nabla f_i(\hat{x}) + \sum \nu_j \nabla h_j(\hat{x}) = 0$$

Constrained convex optimization — 4/4

KKT conditions are a "witness" or "certificate" or "proof" that $\hat{x}$ is optimal.